Performance Monitoring

Performance Unit – Daily Profile

- Global-Forecasting Models (GFM) that learn patterns across many time series
- Each time series differs in scale and behavior
- Performance evaluation:
  - Shared nature of the models,
  - The heterogeneity/dissimilarity across serials.
- Forecast-generation operation: daily per serial
- Evaluation Unit: **single serial forecast on a given forecast day (Serial-Day)**

s — serial

d — forecast day

Performance metric is defined as a function **M:**

$$\mathbf{M(s,d) = M(}A(s,d), F(s,d)\mathbf{)}$$

A(s,d): true measurements

F(s,d): forecasted measurements

A and F will be calculated for all 15-miniute horizons h:

$$h = 1, \ldots, 96$$

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Tracking

Serial

- Daily Performance
- Rolling Performance over a window of D days
  - Trend: MA, EMA
  - Volatility
- **Reference / threshold:**
  - Compare **daily metric** and **rolling performance** against a defined performance target, baseline, or threshold where applicable.
  - Significant deviations can trigger further investigation or adjustment.
- Python based Dashboard/Device Tool

Fleet

- Aggregated Performance (model?!)
  - Calculate metric per serial-day
  - Aggregate per fleet, e.g., average, median
  - Advantage: volatility, distribution
  - Interpretation: **performance of the typical serial**
- Pooled Performance (business?!)
  - Calculate metric per day for all serials
  - Interpretation: **forecast accuracy of the total fleet volume**
- **Scheduled Job**

Performance Metrics

wMAPE	$\Sigma_h \lvert a-f \rvert / \Sigma_h \lvert a \rvert$	Normalized	Error is expressed **relative** to the **actual total volume**
NRMSE — mean actual	$RMSE / mean_h(\lvert a \rvert)$	Normalized	Error is measured **relative** to the **actual mean volume**
NRMSE — sd	$RMSE / SD_h(a)$	Normalized	Error is measured **relative** to the **actual mean variability**
MAPE	$(1/96) \times \Sigma_h \lvert a-f \rvert / \lvert a \rvert$	Normalized	Error is expressed as the average **relative** errors **over horizons**
NRMSE — range	$RMSE / (max_h(a) - min_h(a))$	Normalized	Error is measured **relative** to the **observed range** of **actual values**
RMSE	$\sqrt{[(1/96) \times \Sigma_h (a-f)^2]}$	Dependent	Measures **absolute** error and penalizes large errors more strongly.
MAE	$(1/96) \times \Sigma_h \lvert a-f \rvert$	Dependent	Simple average **absolute** error. Easy to interpret directly in the target's unit.

Performance Metrics – Normalized vs Dependent

	Serial level	Fleet level
Dependent	• **absolute error per serial per day** • Answers: "How many kWh was this serial's forecast wrong by?" • The **actual energy error** of an **individual serial** • Not directly comparable across serials and days	• **absolute error for the whole fleet per day** • Answers: "How many kWh was the fleet forecast wrong by?" • Represents the **total energy impact** of forecasting errors. • Large serials contribute more to the result.
Normalized	• **error relative to the serial's own scale per day**. • Answers: "How large was the error compared with this serial's actual energy?" • Makes performance comparable across serials of different sizes • Better for identifying whether an individual serial is performing poorly relative to its own scale.	• **overall forecasting quality across the fleet per day**, with serials contributing on a comparable basis. • Answers: "How well does the model perform across the fleet, independent of serial size?" • Prevents large serials from dominating the fleet KPI. • Better represents **fleet-wide model performance** rather than total energy impact.

Thanks for your attention!